Distribution Theory I

STA 312

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Outline

1. Introduction
   - Overview
   - Prerequisites
   - Recaps
2. Probability distributions of functions of random variables
   - Derivation methods
   - Implementation Steps 1
   - Implementation Steps 2
   - Implementation Steps 3

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Module Information

- Module Title: Distribution Theory I (3 Units C)
- Module Code: STA 312
- Lecture Days and Times: Wednesday 9 - 11am & Thursday 2 - 3pm.
- Lecture Venue: Statistics Hall 2 & Statistics Lab (for R sessions)
- Assessment: 1 Homework, 1 In-class test, Examination
- Learning resources/further readings: Lecture slides/Notes; Relevant textbooks and online resources.

Specific Learning Objectives

What's here for you?

At the of this module, you should be able to:

- Understand various functions of random variables
- Find the probability distribution of the functions of random variables using:
  - Cumulative distribution function or the method of distribution functions.
  - moment-generating function
  - Variable transformation.
- Explain cumulants and their generating functions.
- Explain central limit theorem (CLT).
- Explain bivariate MGFs and bivariate normal distributions.

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Prerequisites

Required Background knowledge

To successfully achieve all the objectives of this course, you need to have basic knowledge in:

- Probability Theory.
- Types and properties of discrete and continuous probability distributions.
- Relationships between cumulative (density) function $F_X(x)/P_X(x)$ and probability density/mass functions $f(x)/p(x)$.
- Variable transformation methods.
- How to derive moment generation functions (MGFs) of distributions.

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Some common probability distributions:

Discrete probability distributions ($P_x(x)$, $p(x)$)

- Discrete Uniform Distribution
- Bernoulli distribution
- Binomial distribution
- Geometric distribution
- Hypergeometric distribution
- Negative Binomial distribution
- Poisson distribution

Some common probability distributions:

Continuous probability distributions ($F_x(x)$, $f(x)$)

- Uniform Probability distribution
- Gamma probability distribution
- Beta probability distribution
- Exponential probability distribution
- Normal distribution
- Chi-square distribution

Drawing samples from probability distributions is R

We can draw samples from the probability distributions below in R:

- Binomial distribution: rbinom(10000, 5, 0.45)
- Poisson distribution : rpois(10000, 5)
- Gamma probability distribution: gamma(10000, shape = 2, rate = 4)
- Normal distribution: : rnorm(10000, 3, 4)
- Exponential probability distribution: try this?
- Chi-square distribution: : try this?
- : Beta probability distribution : try again?

Moment Generating Function

Moment Generating Function (mgf), $M_x(t)$ is a unique function that completely determines the distribution of the random variable X.

Definition:
$M_x(t) = E(e^{tx})$

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Three (3) key methods:

- Distribution functions: Find the distribution function of the random variables $X_1, \ldots, X_n$, $F(y) = P(Y \leq y)$ by integrating over the regions of X's for which $Y \leq y$ and obtain
$$f(y) = dF(y)/dy.$$
- Transformation of variables:
- Moment-generating functions:

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Distribution function method

Given a sequence of independent random variables $X_1, X_2, \ldots, X_n$ with known probability density function $f(x)$, we are want to obtain the probability distribution function of $Y = h(X)$.

- Set $F(y) = P(Y \leq y)$
- Set $F(y) = P(X \leq h^{-1}(Y))$, where $h^{-1}(Y)$ is the inverse function of Y
- Obtain $F(y)$ by integrating $f(x)$ over the range of values of Xs in terms of y
- Set the range of values of Y based on the relationship with values of X
- Obtain the probability density function of Y, $f(y)$ by differentiating $F(y)$ with respect to y.

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Transformation method

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- Set $F(y) = P(Y \leq y)$
- Set $P(Y \leq y) = P(X \leq h^{-1}(Y))$, where $h^{-1}(Y)$ is the inverse function of Y
- Obtain the derivatives of x with respect to y, $\frac{dx}{dy}$
- Obtain $g(y) = f(x)\frac{dx}{dy}$, where $f(x)$ is evaluated in terms of y

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- Some class exercises ....